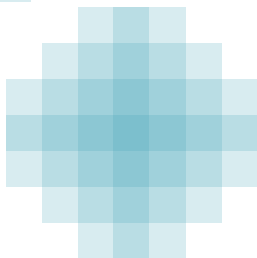
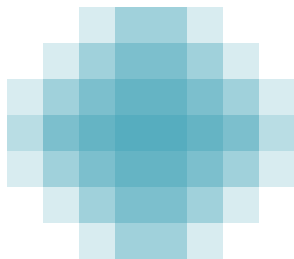


need derivatives with respect to histogram a





Jacobian of $a(\theta)$ wrt θ

Dual potential from EOT between $a(\theta)$ and fixed β

Derivatives w.r.t. Histograms

- For Eulerian discretization, **need derivatives with respect to histogram $\mathbf{a}$**
- For $\varepsilon > 0$, the EOT loss $\text{OT}_c^\varepsilon(\mathbf{a}, \mathbf{b})$ is convex and differentiable w.r.t. $\mathbf{a}, \mathbf{b}$
- Derivatives are given by the optimal dual potentials:

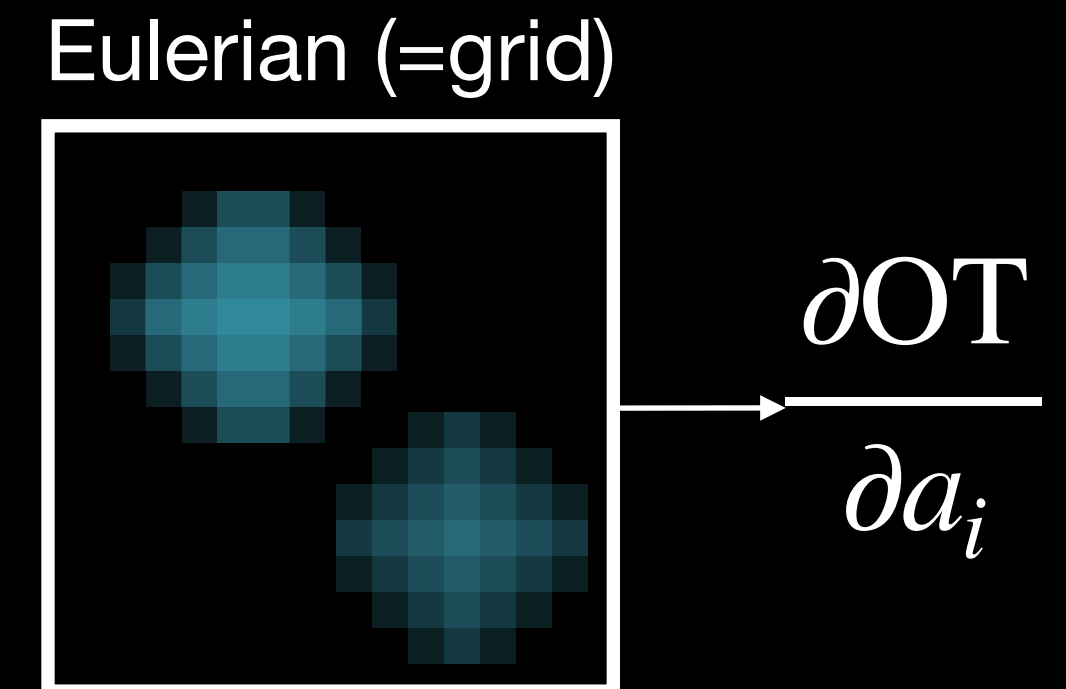
$$\nabla_{\mathbf{a}} \text{OT}_c^\varepsilon(\mathbf{a}, \mathbf{b}) = \mathbf{f} \quad \text{and} \quad \nabla_{\mathbf{b}} \text{OT}_c^\varepsilon(\mathbf{a}, \mathbf{b}) = \mathbf{g}$$

where $(\mathbf{f}, \mathbf{g})$ solve the dual problem, and are zero-mean centered: $\sum_i f_i = \sum_j g_j = 0$

- These dual variables can be used to compute parameter gradients via chain rule:

$$\nabla \mathcal{L}(\theta) = \underbrace{[D_\theta a(\theta)]^\top}_{\text{Jacobian of } a(\theta) \text{ wrt } \theta} \underbrace{\mathbf{f}}_{\text{Dual potential from EOT between } a(\theta) \text{ and fixed } \beta}$$

- Enables gradient-based learning in models where histograms $a(\theta)$ depend on parameters θ



Derivatives w.r.t. Cost and Support

Lagrangian (=particles)

